D10. Evaluation Results and Error Analysis

Quantitative outcomes, difficult cases and truthful interpretation.

Field	Value
Project	Development of AI-Based Cyber Security Audit Assessment Against Benchmark
Scope	Objective 3; secure local/air-gapped RAG proof of concept
Report date	28 August 2026
Evidence rule	Implemented, tested and untested claims are labelled separately.

Observed error taxonomy

Failure pattern	Observed evidence	Response
No interpreted as Yes	3 of 4 No labels missed in the Qwen3 real-report run	Add negative and contradiction-focused labels
Missing-evidence safety unmeasured	No Insufficient Evidence gold examples	Create independently labelled abstention cases
Weak evidence ranking	Recall@8=87.50%; MRR=0.2781 in Qwen3 run	Inspect query/chunk evidence and retriever variants
Malformed structured output	Contextual and large-window conditions produced parse/review flags	Retain schema checks and avoid unvalidated settings

Interpretation of the headline result

The 81.25% Qwen3 accuracy is a useful POC result because it comes from a real executed two-report workflow, not an illustrative demo. It is not enough to claim reliable assessment across unseen agencies, report formats or missing-evidence cases. [E4][E5][E10]

The architecture's value is that these limitations are visible. The harness surfaces ranking metrics, response errors, flags and class imbalance instead of presenting only the headline audit score. [E5][E10]

Primary result: two-report JSON proof set

Measure	Observed value	Meaning
Report-question pairs / chunks	16 / 22	Small integration dataset [E4]
Answer accuracy / macro-F1	81.25% / 0.4296	Macro-F1 limited by No errors and zero abstention support [E5]
Precision@8 / Recall@8	14.43% / 87.50%	Evidence is often found but weakly ranked [E5]
MRR / nDCG@8	0.2781 / 0.4212	Relevant evidence is not consistently near the top [E5]
Grounding / CRAG-correct	100% / 25.00%	Citation-boundary pass is not semantic entailment [E5][E8]
CRAG reformulation / mean latency	68.75% / 13.66 s	Retrieval judge is frequently uncertain; end-to-end latency is measurable [E5]

Audit-level output

Report	Positive / total	Score	Decision
2-REPORT	8 / 8	100.00%	PASS at 70% threshold [E5]
4-REPORT	7 / 8	87.50%	PASS at 70% threshold [E5]

Error analysis

The answer confusion matrix contains 12 correct Yes responses, one correct No response and three No-to-Yes errors. This makes universal or negative report statements the highest-priority failure type. [E5]

Two queries have zero recall@8. Raising the context limit alone would conceal the diagnosis; inspect gold labels, chunk boundaries and query formulation, then run isolated retrieval ablations. [E5]

The grounder checks evidence-boundary/lexical support, so 100% grounding should not be reported as proof that every answer is entailed by the cited passage. [E5][E8]

Related saved results

The separate Qwen3 retrieval comparison tied B0 on a very small one-report corpus. The Qwen3-0.6B end-to-end run achieved 58.33% accuracy and predicted No for all questions, showing that a newer/smaller generator is not automatically a better choice. [E2][E3]

The two-report proof is a useful integration result but not a production reliability study. [E4][E5]

Priority actions

Add independently reviewed No and Insufficient Evidence labels; calibrate CRAG; compare chunk/retrieval variants under a frozen test; evaluate claim entailment and contradiction handling.

Observed Qwen3 decision errors

Gold class	Predicted Yes	Predicted No	Interpretation
Yes (n=12)	12	0	The run captured all source-derived positive labels.
No (n=4)	3	1	Negative claims are the primary observed failure mode.
Insufficient Evidence (n=0)	0	0	Abstention quality cannot be estimated from this set.

Error-analysis conclusion

The 81.25% headline accuracy must be read together with the confusion matrix. The model's 100% recall on Yes is encouraging for this POC, but No recall is only 25%; the system tends to convert absence/negative statements into a positive conclusion. Macro-F1 of 0.4296 captures this imbalance better than accuracy alone. [E5]

The correct response is not to hide the failure in an aggregate score. The next dataset should deliberately include negative, contradiction and missing-evidence questions with independent labels, while retrieval and prompt changes are tested against that frozen set. [E10]

Prioritised remediation plan

Priority one is negative and absence-sensitive reasoning. The three false-positive No cases show that the generator needs evidence that positively establishes a universal claim such as all findings being closed; the absence of a contradiction must not become a Yes. This should be addressed through new labels, prompt tests and claim-level verification. [E5][E8]

Priority two is retrieval ordering. With MRR 0.2781, the relevant evidence is often available but buried below competing content. Reranking should be tested as one isolated variable on a re-labelled evidence benchmark, rather than enabled as an unmeasured architecture embellishment. [E5][E10]

Priority three is abstention. The schema supports Insufficient Evidence, but its safety cannot be measured until that class exists in the gold data. Completion of this data work is the release-gate requirement; it is not something an attractive accuracy score can substitute. [E10]

Repository evidence cited

- [E4] results/production_real_reports/1787846335_de0ded86a7db/run_config.json and run.log: two-report production-parser run configuration and execution log.
- [E5] results/production_real_reports/1787846335_de0ded86a7db/{retrieval_metrics,question_level_llm_metrics,sota_faithfulness_metrics,audit_level_scores}.json: scored metrics and audit decisions.
- [E10] results/final_validation/VALIDATION_SUMMARY.md and its cited run directories: controlled local LLM, ICL, prompt, retrieval, CRAG, chunking and chunk-parameter comparisons completed 28 August 2026.